

HOMEWORK 3

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 1.5 - BASIC PROPERTIES OF ANALYTIC FUNCTIONS

1. On what sets are the following functions analytic? Compute the derivative for each.

(a) z^n for $n \in \mathbb{Z}$

We consider three cases: First, when $n = 0$, $z^n = z^0 = 1$ is constant, hence it is analytic on $\mathbb{C}$ and $\frac{d}{dz}z^n = 0$. Second, when $n > 0$, proposition 1.5.3 part (iv) yields that z^n is analytic on $\mathbb{C}$ and $\frac{d}{dz}z^n = nz^{n-1}$. Finally, when $n < 0$, Proposition 1.5.3 part (v) yields that z^n is analytic on $\mathbb{C} \setminus \{0\}$ and, combining Proposition 1.5.3 part (iii) and Proposition 1.5.3 part (iv),

$$\frac{d}{dz}z^n = \frac{d}{dz} \frac{1}{z^{|n|}} = \frac{(0)(z^{|n|}) - (1)(|n|z^{|n|-1})}{(z^{|n|})^2} = \frac{nz^{|n|-1}}{z^{2|n|}} = nz^{-|n|-1} = nz^{n-1}.$$

(b) $\frac{1}{\left(z + \frac{1}{z}\right)^2}$

First, note that $\frac{1}{\left(z + \frac{1}{z}\right)^2} = \left(z + \frac{1}{z}\right)^{-2} = \left(\frac{z^2 + 1}{z}\right)^{-2} = \frac{z^2}{(z^2 + 1)^2}$. By Proposition 1.5.3 part (v), this function is analytic on $\mathbb{C} \setminus \{i, -i\}$ and, combining Proposition 1.5.3 part (iii), Proposition 1.5.3 part (iv), and Theorem 1.5.4, we have

$$\begin{aligned} \frac{d}{dz} \left(\frac{1}{\left(z + \frac{1}{z}\right)^2} \right) &= \frac{d}{dz} \left(\frac{z^2}{(z^2 + 1)^2} \right) \\ &= \frac{(z^2 + 1)^2 \frac{d}{dz}(z^2) - (z^2) \frac{d}{dz}((z^2 + 1)^2)}{\left((z^2 + 1)^2\right)^2} \\ &= \frac{(z^2 + 1)^2(2z) - (z^2)(2(z^2 + 1)(2z))}{(z^2 + 1)^4} \\ &= \frac{(z^2 + 1)(2z)((z^2 + 1) - 2(z^2))}{(z^2 + 1)^4} \\ &= \frac{2z(1 - z^2)}{(z^2 + 1)^3}. \end{aligned}$$

(c) $\frac{z}{z^n - 2}$ for $n > 0, n \in \mathbb{Z}$

We consider two cases: First, when $n = 0$, $\frac{z}{z^n - 2} = \frac{z}{1 - 2} = -z$ and by proposition 1.5.3 part (iv), $\frac{d}{dz} - z = -1$. Second, when $n > 0$, then by Proposition 1.5.3 part (v), the function is analytic everywhere except the n^{th} roots of 2 and, combining Proposition 1.5.3 part (iii) and Proposition 1.5.3 part (iv)

$$\begin{aligned}\frac{d}{dz} \left(\frac{z}{z^n - 2} \right) &= \frac{(z^n - 2) \frac{d}{dz}(z) - (z) \frac{d}{dz}(z^n - 2)}{(z^n - 2)^2} \\ &= \frac{(z^n - 2)(1) - (z)(nz^{n-1})}{(z^n - 2)^2} \\ &= \frac{-(n-1)z^n - 2}{(z^n - 2)^2}.\end{aligned}$$

2. Prove that $\frac{df^{-1}}{dw} = \frac{1}{f'(z)}$ where $w = f(z)$ by differentiating $f^{-1}(f(z)) = z$ using the chain rule.

Assume that f^{-1} is defined and is analytic.

Theorem 1.4.5 (i.e. the chain rule) states that if $A, B \subseteq \mathbb{C}$ are open sets and $f : A \rightarrow \mathbb{C}, g : B \rightarrow \mathbb{C}$ are analytic functions such that $f(A) \subseteq B$, then $g \circ f : A \rightarrow \mathbb{C}$ is analytic and $(g \circ f)'(z) = g'(f(z))f'(z)$. Accordingly, if $g = f^{-1}$ is defined and analytic, then certainly $f(A) \subseteq B = \text{dom } f^{-1}$, and

$$(f^{-1} \circ f)'(z) = (f^{-1})'(f(z))f'(z).$$

Written differently, if $w = f(z)$, then $f^{-1}(w) = z$ and we have

$$\begin{aligned}\frac{d}{dz}(f^{-1}(w)) &= \frac{d}{dz}(z) \\ \frac{df^{-1}}{dw} \frac{dw}{dz} &= 1 \\ \frac{df^{-1}}{dw} \frac{df}{dz} &= 1 \\ \frac{df^{-1}}{dw} &= \frac{1}{f'(z)}.\end{aligned}$$

3. Use the inverse function theorem to show that if $f : A \rightarrow \mathbb{C}$ is analytic and $f'(z) \neq 0$ for all $z \in A$, then f maps open sets in A to open sets.

To begin, recall that Theorem 1.5.10 (i.e. the inverse function theorem) tells us that if $f : A \rightarrow \mathbb{C}$ is analytic and $f'(z_0) \neq 0$, then there exists a neighborhood U of z_0 and a neighborhood V of $f(z_0)$ such that $f : U \rightarrow V$ is a bijection and its inverse function $f^{-1} : V \rightarrow U$ is analytic with derivative $\frac{d}{dw}f^{-1}(w) = \frac{1}{f'(z)}$, where $w = f(z)$. Suppose $f'(z) \neq 0$ for all $z \in A$. Let $\mathcal{O} \subseteq A$ be an open set and note that the restriction $f|_{\mathcal{O}} : \mathcal{O} \rightarrow \mathbb{C}$ satisfies $f|_{\mathcal{O}}'(z) = f'(z) \neq 0$ for all $z \in \mathcal{O}$. So, we need only show that $f(A)$ is open, since we can then show the same for $f(\mathcal{O})$ by restricting f to $\mathcal{O}$. Accordingly, near any point $z \in A$, there exists a neighborhood U_z of z and V_z of $f(z)$ such that $f : U_z \rightarrow V_z$ is invertible and has an analytic inverse. Hence, $f(U_z) = V_z$, and since some such set $U_z \subseteq A$ exists for all $z \in A$, we have

$$f(A) = f\left(\bigcup_{z \in A} U_z\right) = \bigcup_{z \in A} f(U_z) = \bigcup_{z \in A} V_z.$$

Since V_z is open, $\bigcup_{z \in A} V_z$ is a union of open sets and is open as well. Therefore, $f(A)$ is open and the result follows.

4. Define $\frac{\partial f}{\partial \bar{z}}$ by $\frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} - \frac{1}{i} \frac{\partial f}{\partial y} \right)$. Show that the Cauchy-Riemann equations are equivalent to $\frac{\partial f}{\partial \bar{z}} = 0$.

Write $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. We will show that the Cauchy-Riemann equations are equivalent to $\frac{\partial f}{\partial \bar{z}} = 0$. To do so, we will first show that having $\frac{\partial f}{\partial \bar{z}} = 0$ implies the Cauchy-Riemann equations. Then, we will show that Cauchy-Riemann equations imply that $\frac{\partial f}{\partial \bar{z}} = 0$.

Assume that $\frac{\partial f}{\partial \bar{z}} = 0$. Then, since $\frac{1}{i} = \frac{1}{i} \begin{pmatrix} -i \\ -i \end{pmatrix} = -i$, it follows that

$$\begin{aligned} \frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} &= 0 \\ \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} + i \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) &= 0 \\ \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} + i \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} &= 0 \\ \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) + i \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) &= 0. \end{aligned}$$

Accordingly, it must be the case that

$$\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0 \text{ and } \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} = 0,$$

which is equivalent to

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

Therefore, $\frac{\partial f}{\partial \bar{z}} = 0$ implies that f satisfies the Cauchy-Riemann equations.

In the reverse direction, suppose f satisfies the Cauchy-Riemann equations. Then, we have

$$\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) + i \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) = 0$$

and, by the equalities established in the forward direction, the result follows.

5. On what sets are each of the following functions harmonic?

(a) $u(x, y) = \operatorname{Im}\left(z + \frac{1}{z}\right)$ (where $z = x + iy$)

Note that $\operatorname{Im}\left(z + \frac{1}{z}\right) = \operatorname{Im}\left(\frac{z^2 + 1}{z}\right)$. By Proposition 1.5.12, if f is analytic on an open set A , then $\operatorname{Re} f$ and $\operatorname{Im} f$ are harmonic on A . Since $\frac{z^2 + 1}{z}$ is analytic on $\mathbb{C} \setminus \{0\}$ by Proposition 1.5.3 part (v), it follows that u is harmonic on $\mathbb{C} \setminus \{0\}$ as well.

(b) $u(x, y) = \frac{y}{(x-1)^2 + y^2}$

We proceed by direct computation, noting that u is defined and differentiable for $(x, y) \neq (1, 0)$, producing:

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{-(2(x-1)y)}{((x-1)^2 + y^2)^2} & \frac{\partial u}{\partial y} &= \frac{x^2 - 2x - y^2 + 1}{((x-1)^2 + y^2)^2} \\ \frac{\partial^2 u}{\partial x^2} &= \frac{-(2y(-3x^2 + 6x + y^2 - 3))}{((x-1)^2 + y^2)^3} & \frac{\partial^2 u}{\partial y^2} &= \frac{2y(-3x^2 + 6x + y^2 - 3)}{((x-1)^2 + y^2)^3} \end{aligned}$$

Adding, it is clear that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for all values $(x, y) \in \mathbb{R}^2 \setminus \{(1, 0)\}$